A Project Report on

"TEJAS-PRAHARI"

Submitted in fulfillment of the requirements of
Post Graduate Diploma in Advanced Computing

SUBMITTED BY

BATCH OF AUGUST 2025

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CERTIFICATE

This is to certify that the Report work entitled

"TEJAS-PRAHARI"

Has been duly completed by the following students under the my guidance, in a satisfactory manner as a partial fulfillment of the requirement for the award of the E- Diploma in Advanced Computing, Delhi

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Declaration

I declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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PROJECT TEAM

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ABSTRACT

The rapid expansion of digital education and remote learning, ensuring the security, reliability, and scalability of online examinations has become a critical challenge. The Tejas-prahari is a web-based system designed to overcome the limitations of traditional examination methods, such as manual evaluation, limited transparency, and vulnerability to malpractice. The system provides a secure examination environment through role-based authentication, controlled exam workflows, and automated evaluation mechanisms.

Developed using Java 17 and Spring Boot for the backend, the system exposes robust RESTful APIs consumed by a React-based frontend, enabling seamless interaction between users and the application. MySQL is integrated for efficient and scalable data management, ensuring data consistency and reliability across examination processes. By automating exam conduction, monitoring, and evaluation, the Tejas-prahari enhances operational efficiency, minimizes human intervention, and promotes trust and fairness in digital assessments.

“TEJAS-PRAHARI”
(Online secure examination browser)

Introduction

The rapid advancement of information technology has significantly transformed the education sector. With the widespread adoption of online learning platforms, conducting examinations in a secure, fair, and scalable manner has become a critical challenge. Traditional examination systems, whether paper-based or basic online forms, often suffer from issues such as manual evaluation delays, high administrative costs, lack of scalability, and vulnerability to malpractice.

The **Tejas-prahari** is designed to address these challenges by providing a controlled, secure, and efficient environment for conducting online examinations. The system ensures that students can only access authorized exam content while restricting access to external applications, websites, screen capture tools, and communication software during the examination period.

The proposed system integrates authentication mechanisms, browser lockdown techniques, real-time monitoring, and automated evaluation to maintain exam integrity. By leveraging modern web technologies and security practices, the Tejas-prahari aims to deliver a reliable and tamper-resistant examination experience suitable for educational institutions, certification bodies, and corporate assessments.

Problem Statement

Traditional examination systems face several challenges:

1. Manual exams are time-consuming and costly
Require printing large numbers of question papers and answer sheets.Need physical arrangements such as exam halls, seating plans, and invigilators. Involve administrative effort for distribution, collection, and storage of papers. Increase expenses for paper, ink, transport, and staff management. Preparation and post-exam processing take several days or weeks.

2. Risk of cheating and impersonation
Students may attempt to copy answers from others during exams. Unauthorized materials like notes or electronic devices may be used. Someone else may try to take the exam on behalf of a candidate.Manual monitoring by invigilators may not detect all unfair activities.Lack of automated verification reduces exam integrity.

3. Difficulty in managing large numbers of students
Requires arranging multiple rooms and supervisors.Scheduling exams for many candidates becomes complex.Managing attendance and seating allocation is challenging.Handling question paper distribution for large groups is time-intensive.Administrative workload increases significantly as student numbers grow.

4. Delay in evaluation and result generation
Teachers must manually check each answer sheet. Calculating marks and preparing result sheets takes considerable time. Large batches of students increase correction workload. Human errors in totaling or marking may occur. Results may be delayed, affecting academic decisions and schedules.

Objective

The main objectives of the Tejas-prahari are:

- To provide a secure platform for online exams.
- To enable easy exam creation and management.
- To provide real-time result generation.
- To maintain user authentication and role-based access.
- To provide analytics for performance evaluation.

Objectives – Detailed Explanation

1. To provide a secure platform for online exams

Ensure that only authorized users can access the examination system. Protect exam data such as question papers and student responses from unauthorized access. Maintain confidentiality and integrity of exam content. Prevent leakage of question papers before or during exams. Create a reliable environment where exams can be conducted safely.

2. To automate exam management and evaluation

Automatically schedule and organize examinations within the system. Display questions digitally without manual distribution.Record student responses instantly as they are submitted.Evaluate answers automatically where applicable.Generate results quickly without manual checking.

3. To ensure authentication and fairness

Verify candidate identity before allowing access to the exam.Prevent impersonation or unauthorized participation.Apply time limits uniformly for all candidates.Ensure each student receives the same level of difficulty and instructions.Maintain transparency in the evaluation process.

4. To reduce administrative workload

Eliminate the need for printing and distributing physical papers. Reduce manual effort in arranging exam halls and invigilators. Simplify student data and exam record management. Minimize paperwork and storage requirements. Allow administrators to manage exams efficiently through a centralized system.

Technologies Used

- Programming Language: Java 17
- Backend Framework: Spring Boot
- Frontend Framework: React.js, HTML, CSS
- Database: MySQL
- Build Tool: Maven
- Server: Tomcat

Scope of the Project

The scope of the Tejas-prahari includes managing the entire lifecycle of examinations, from question creation to result declaration. The system is suitable for educational institutions, certification authorities, training centers, and organizations conducting assessments. It supports multiple exam formats such as quizzes, objective tests, and structured assessments.

The platform includes functionalities such as user registration, exam scheduling, question management, monitoring, scoring, and reporting. It is designed to be flexible so that it can be adapted to different examination requirements. The system can also be expanded or customized based on institutional needs, making it a versatile solution for various types of assessments.

Existing System

Existing System:-

In the existing examination system, assessments are conducted either through traditional **offline (paper-based)** methods or through **basic online examination platforms** that operate on standard web browsers. In offline systems, exams require physical infrastructure, manual invigilation, printed question papers, and human evaluation, making the process time-consuming, costly, and prone to errors.

In basic online examination systems, candidates attempt exams using common web browsers such as Chrome or Edge. These systems primarily focus on question delivery and result generation but lack strict security controls. Students can easily switch tabs, open other applications, use external devices, communicate with others, or capture exam content using screen recording tools. Monitoring is often minimal or completely absent, relying heavily on trust rather than enforcement.

As a result, existing systems fail to ensure exam integrity, especially in remote or large-scale online examinations.

Limitation Of the Existing System

The existing system suffers from several limitations:

1. **Lack of Security Controls:-**
   Standard browsers allow tab switching, application switching, and access to unauthorized websites.
2. **High Possibility of Malpractice:-**
   Students can use unfair means such as searching answers online, screen sharing, or communication tools.
3. **No Browser Lockdown:-**
   Existing online platforms do not restrict keyboard shortcuts, screen capture, or external applications.
4. **Limited Monitoring:-**
   Real-time tracking of user activity is either weak or completely absent.
5. **Manual Effort and Errors:-**
   Offline exams require manual checking, which is time-consuming and prone to human error.
6. **Scalability Issues:-**
   Conducting exams for a large number of candidates is difficult and resource-intensive.
7. **Delayed Results:-**
   Manual or semi-automated evaluation leads to delays in result declaration.

Need For the Proposed System

The growing adoption of online education and remote assessments has created a strong need for a **secure, reliable, and scalable examination solution**. The proposed **Tejas-prahari** addresses the shortcomings of existing systems by providing a controlled exam environment that prevents unauthorized activities and ensures fairness.

The need for the proposed system arises due to the following reasons:

1. **Ensuring Exam Integrity:-**
   The system prevents cheating by restricting access to external resources and applications.
2. **Secure Browser Environment:-**
   Browser lockdown mechanisms ensure that students remain within the exam interface throughout the test.
3. **Automation and Efficiency:-**
   Automated evaluation and instant result generation reduce administrative workload.
4. **Scalable Online Examinations:-**
   The system supports large-scale exams without the need for physical infrastructure.
5. **Remote Accessibility:-**
   Enables institutions to conduct exams from any location without compromising security.
6. **Transparency and Fairness:-**
   Activity logs and monitoring improve trust in online assessments.
7. **Cost and Time Savings:-**
   Eliminates printing, logistics, and manual invigilation costs.

Proposed System

The proposed Tejas-prahari is a web-based system combined with browser-level restrictions to ensure exam integrity. The system allows administrators to create and manage exams, while students can attempt exams only through the secure browser interface.

Key Features

- Controlled browser environment
- Secure authentication
- Time-bound exams
- Automated evaluation
- Real-time activity logging
- Centralized exam management

Module Description

1. User Authentication Module

This module ensures that only authorized users can access the examination system. It verifies credentials and assigns roles such as student or administrator.

2 Admin Module

The admin module allows administrators to:

- Create exams
- Add questions
- Set time limits
- Manage users
- View results and logs

3 Exam Module

This module presents questions to students in a secure interface. It enforces time limits and ensures auto-submission upon exam completion.

4 Secure Browser Module

This is the core component of the system. It:

- Disables tab switching
- Blocks external websites
- Prevents screen capture
- Restricts keyboard shortcuts

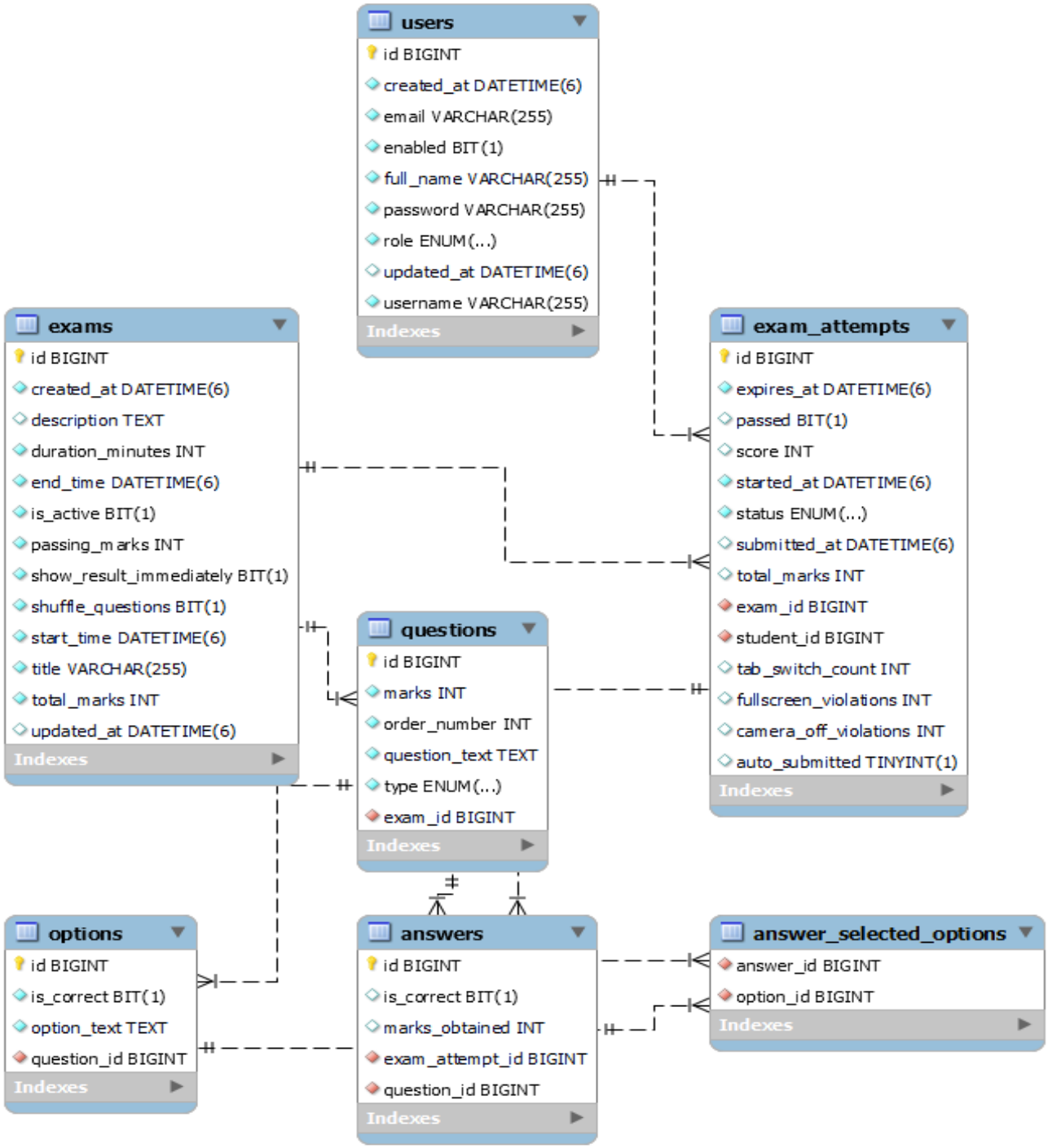
5 Evaluation Module

Automatically evaluates objective-type questions and generates results instantly after submission.

Advantage of this System

- Prevents cheating effectively
- Reduces administrative workload
- Saves time and cost
- Instant evaluation and results
- Scalable for large examinations
- Improves exam credibility

ER Diagram:-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enabled BIT(1)
full_name VARCHAR(255)
password VARCHAR(255)
role ENUM(...)
updated_at DATETIME(6)
username VARCHAR(255)
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is_active BIT(1)
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question_id BIGINT
Indexes
answer_selected_options
answer_id BIGINT
option_id BIGINT
Indexes

Limitations of the System

Despite its advantages, the system has certain limitations. It requires a stable internet connection for smooth operation, which may not always be available in remote areas. Users must possess basic digital skills to interact with the platform effectively. Technical issues such as system downtime or device failure may disrupt exam sessions.

Additionally, the system depends on continuous availability and proper maintenance to function efficiently. These limitations highlight the importance of technical support, infrastructure readiness, and user training for successful implementation.

Future Enhancements

Future improvements can further strengthen the system's capabilities. Advanced monitoring features can enhance exam supervision. Improved analytics and reporting tools can provide deeper insights into candidate performance. Integration with other academic platforms can streamline administrative processes.

Enhancements in user interface design can improve usability and user experience. Additional assessment formats can expand system functionality. Continuous development and updates will ensure that the system remains relevant and effective in evolving educational environments.

Applications of the System

The system can be used in various contexts, including academic examinations, competitive tests, certification programs, recruitment assessments, and training evaluations. Educational institutions can use it for internal exams, while organizations can use it for skill assessments and selection processes. Its adaptability makes it suitable for multiple sectors requiring reliable evaluation methods.

Conclusion

The Tejas-prahari provides a comprehensive solution for conducting examinations in a secure, efficient, and reliable digital environment.

By automating exam processes, it reduces administrative burden, enhances accuracy, and ensures fairness. The system addresses challenges associated with traditional examinations while supporting modern educational requirements.

Its structured design, security measures, and scalability make it a valuable tool for institutions and organizations seeking to modernize assessment methods. As digital transformation continues to shape education and professional evaluation, such systems will play a vital role in ensuring efficient and credible examinations.

Bibliography

- Academic and institutional examination guidelines
- Standard assessment procedures
- Principles of educational evaluation
- System design and documentation resources